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Date: January 2025

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Corrigendum to “Wettability and interfacial reactions in Al–Mg–Si/SiC and Al–Mg–Si/SiO₂ systems” Materials Characterization 219 (2025) 114599

Yu-Jie Ge, ..., Ping Shen

Materials Characterization, Volume 233, March 2026, Pages 116054

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Outline

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Highlights

- Wetting and spreading were measured in Ar–10%H₂ using a dispensed sessile drop method.
- Lower Si and higher Mg in Al–Mg–Si alloy improve wettability of SiC.

- Increased Mg content promotes MgAl₂O₄ formation at Al–Mg–Si/SiO₂ interface.
- SiC pre-oxidation controls Al₄C₃ formation rather than directly enhancing wettability.
- A spherical cap model corrects contact angle for infiltration-induced deviations.

Abstract

The wettability of reinforcing materials by matrix alloy is essential for the successful fabrication of high-performance metal matrix composites. This study investigates the wetting behavior of Al–Mg–Si alloys (Mg: 3–7 wt%, Si: 9–15 wt%) on SiC and SiO₂ substrates, providing crucial insights for developing SiC-reinforced aluminum matrix composites. Using a dispensed sessile drop method in an Ar–10%H₂ atmosphere, we systematically examined the influence of alloy composition and temperature on wettability. A key finding is that reducing Si content and increasing temperature significantly enhance wettability in the Al–Mg–Si/SiC system by promoting the formation of Al₄C₃. In contrast, for the Al–Mg–Si/SiO₂ system, increasing Mg content primarily improves initial wettability, while the effect of Si content is less significant. Importantly, this study challenges the conventional understanding of SiC pre-oxidation treatment, demonstrating that its primary role is to control Al₄C₃ formation rather than directly enhance wettability. These results provide a deeper understanding of the complex interplay among alloying elements, temperature, and interfacial reactions in governing wettability. Furthermore, they offer valuable strategies for process optimization and theoretical guidance for fabricating SiC-reinforced aluminum matrix composites with enhanced properties.

Keywords

[Declaration of competing interest](#)

[Acknowledgements](#)

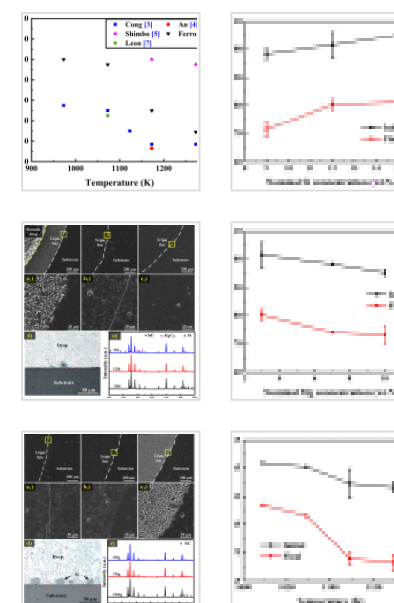
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1. Introduction

Silicon carbide (SiC) reinforced aluminum (Al) matrix composites (AMCs) are increasingly utilized in demanding fields such as aerospace and electronic packaging due to their combination of high thermal conductivity, stiffness, low density, and relative low cost [1,2]. These composites are typically fabricated using liquid-phase techniques like stir casting and infiltration. However, the success of these manufacturing processes and the overall performance of the resulting AMCs are heavily influenced by the wettability and reactivity between the molten aluminum alloy and the SiC reinforcement. Therefore, a comprehensive understanding of the interfacial phenomena governing the interaction between Al and its alloys with SiC is crucial.

Wettability, often quantified by the contact angle (θ) as defined by Young's equation (Eq. (1)), measures the tendency of a liquid to spread on a solid surface,

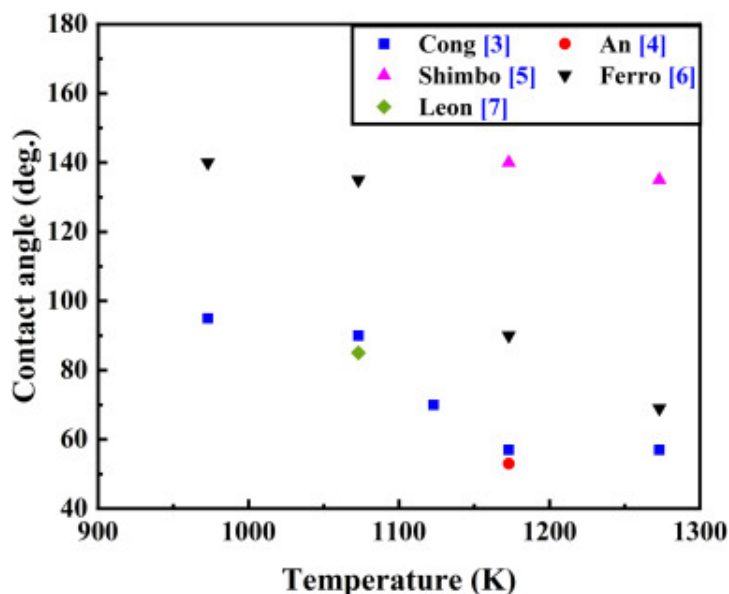
$$\cos\theta = \sigma_{sv} - \sigma_{sl} / \sigma_{lv}, \quad (1)$$

where σ_{sv} , σ_{sl} , and σ_{lv} represent the interfacial energies of the solid/vapor, solid/liquid, and liquid/vapor interfaces, respectively. Additionally, the work of adhesion (W_{ad}) is defined by

$$W_{ad} = \sigma_{lv} \cos \theta + 1, \quad (2)$$

which quantifies the strength of the interfacial bond between the liquid and the solid phase.

Despite extensive research on the wettability of SiC by molten Al [[3], [4], [5], [6], [7]], significant scatter in reported contact angle data (see Fig. 1) has hindered a definitive understanding of the true wettability in this system. This variability is largely attributed to the presence of a native oxide layer (Al₂O₃) on the aluminum surface [8], which acts as a barrier, preventing direct Al–SiC contact and affecting the measured contact angle. Eliminating this oxide layer is challenging, as achieving the extremely low oxygen partial pressure (below 10⁻⁴⁴Pa) required to prevent oxidation at aluminum's melting point is practically impossible [9]. Although high temperatures and vacuum conditions can facilitate the removal of the alumina layer through a self-cleaning reaction ($4\text{Al}_l + \text{Al}_2\text{O}_{3s} = 3\text{Al}_2\text{O}_{g}$ [10]), these conditions also promote the formation of Al₄C₃, a brittle and easily hydrolyzed compound that degrades the composite's performance.



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Fig. 1. Collected equilibrium contact angles for the Al/SiC system as a function of temperature [[3], [4], [5], [6], [7]].

To improve the wettability of the Al/SiC system and suppress Al₄C₃ formation, several strategies have been explored, including adjusting process temperatures and adding alloying elements. For example, Cong et al. [11] demonstrated that Al₄C₃ formation could be avoided below 1173 K for the Al–10 wt%Si/SiC system. However, the effect of silicon on wettability remains controversial. Laurent et al. [12] found that silicon addition had little influence on the wetting behavior of Al–Si alloys on single-crystal SiC, while Wang et al. [13] observed improved wettability with increasing silicon content. This discrepancy underscores the need for further investigation into the role of silicon in the Al/SiC interface. Similarly, the addition of magnesium (Mg) to aluminum has been explored as a way to improve wettability. Liu and Lin [14] reported increasing Mg content (Mg=0–12 wt%) in the Al–Mg/SiC system at 973–1273 K improved wettability due to the deoxidation effect of Mg, which also reduced the interfacial reaction between Al and SiC. However, thermodynamic calculations by Shi et al. [15] contradicted these findings, predicting that Al₄C₃ would persist unless the Mg content exceeded 94.7 mol%. These conflicting results highlight the complex effects of alloying elements on the wettability and interfacial reactions in the Al alloy/SiC system.

Pre-oxidation of SiC particles, resulting in a SiO₂ surface layer, is another common strategy for improving the performance of SiC-reinforced AMCs [[16], [17], [18]]. The SiO₂ layer is believed to hinder Al₄C₃ formation and enhance wettability [[18], [19], [20]]. For instance, Bardal [20] demonstrated the beneficial effect of pre-oxidation on the infiltration of Al–0.05Mg–6Si and Al–0.6Mg–6Si alloy melts into SiC and SiC_{ox} powders at 1373 K by examining the threshold pressure. However, direct comparisons of the wettability of SiC and SiO₂ by molten Al or Al alloys under identical conditions are scarce, making it difficult to fully understand the role of the oxide layer in influencing wettability.

In our previous study, we investigated the wetting behavior of molten Al and its alloys on both pristine and pre-oxidized SiC (denoted as SiC_{ox}) substrates. Contrary to expectations, pre-oxidation deteriorated rather than enhanced wettability (unpublished results; see supplementary material). This was attributed to the instability of the SiO₂ layer formed

on the SiC surface, which readily facilitated the oxidation of the Al droplet, resulting in a large contact angle. Consequently, employing a stable SiO₂ substrate is essential for accurately evaluating the wettability in this system.

Furthermore, the existing literature presents a complex and often conflicting understanding of the interfacial reactions between aluminum alloys and SiO₂. For instance, Tsunekawa et al. [21] studied the Al–3at.%Mg/SiO₂ system at 1073K and identified Al₂O₃ and MgAl₂O₄ as the primary interfacial reaction products. In contrast, Stephenson et al. [22], investigating the same system within a temperature range of 943–1073K, reported MgAl₂O₄ as the sole interfacial product, with MgO formation observed only when the Mg content was increased to 10at.%. Tsujikawa et al. [23] examined the temperature dependence of the reaction between liquid aluminum, Al–5wt%Mg alloy, and silica. They observed a maximum reaction rate around 1173K, which decreased by an order of magnitude at 1273K. The reaction products formed above this peak temperature exhibited a coarser microstructure compared to those formed below, with an intermediate layer present between the reacted and unreacted SiO₂. This decrease in reaction rate above 1173K was attributed to the reduced diffusion rate of Al in the intermediate layer. Hoshino et al. [24] concurred with this observation and suggested that the intermediate layer might be mullite, although conclusive evidence was lacking.

Given these discrepancies and the incomplete understanding of the wettability and interfacial reactions of aluminum alloys on both SiC and SiO₂, this study investigates the wetting behavior and interfacial characteristics of Al–Mg–Si alloys on both substrates using a dispensed sessile drop method under an Ar–10%H₂ atmosphere. Our objective is to elucidate the underlying wetting mechanisms and interfacial chemistry, clarifying the influence of alloying elements and temperature. These findings will provide a theoretical framework and practical guidance for the fabrication of high-performance SiC-reinforced AMCs.

2. Materials and methods

Polycrystalline α -SiC substrates (98.5 wt% purity, 20 mm diameter, 5 mm thickness) were prepared by pressureless sintering, achieving a relative density greater than 97%. The substrates were mechanically polished using diamond paste to an average surface roughness of approximately 20 nm. High-purity (99.99%) quartz substrates (20 mm diameter, 2 mm thickness, <5 nm surface roughness) were used as the SiO₂ substrates. The metallic material employed was an Al–Mg–Si alloy, with Mg content ranging from 3 to 10 wt% and Si content between 9 and 15 wt%. Alloy ingots were sectioned into approximately 0.1 g pieces using wire electrical discharge machining. Both the substrate and alloy piece were ultrasonically cleaned in acetone three times.

Wetting experiments were conducted using the dispensed sessile drop method [9] under isothermal conditions ranging from 1023 to 1173 K in an Ar–10% H₂ atmosphere. Cleaned SiC or SiO₂ substrates were placed and leveled on an alumina pillar within a stainless-steel furnace chamber. The aluminum alloy piece was held in a stainless-steel bellows located outside the chamber and connected to an alumina dispensing tube with a 1 mm diameter orifice at its lower end. The bottom of the alumina tube was positioned at the center of the furnace's heating zone, approximately 5 mm above the substrate. A schematic diagram of experimental device can be found in Ref. [25].

Each experiment began with evacuating the chamber to approximately 5×10^{-4} Pa and heating it under vacuum to 100 K above the target wetting temperature at a rate of 20 K/min. This temperature was maintained for 15 min to desorb surface impurities. The furnace was then cooled to the target wetting temperature at 10 K/min and held isothermally. Upon reaching a vacuum level of 5×10^{-4} Pa, the molecular pump was deactivated, and dehydrated and deoxidized Ar–10% H₂ gas (both Ar and H₂ with purity exceeding 99.999%) was rapidly introduced. When the chamber pressure reached 0.12 MPa, the aluminum alloy piece was released from the bellows into the alumina dispensing tube. Upon complete melting, the pressure difference forced the molten alloy through the tube orifice, forming a droplet on the substrate and simultaneously removing the oxide film from the alloy surface. A high-resolution charge-coupled device camera captured images of the droplet, with the moment of detachment from the alumina tube

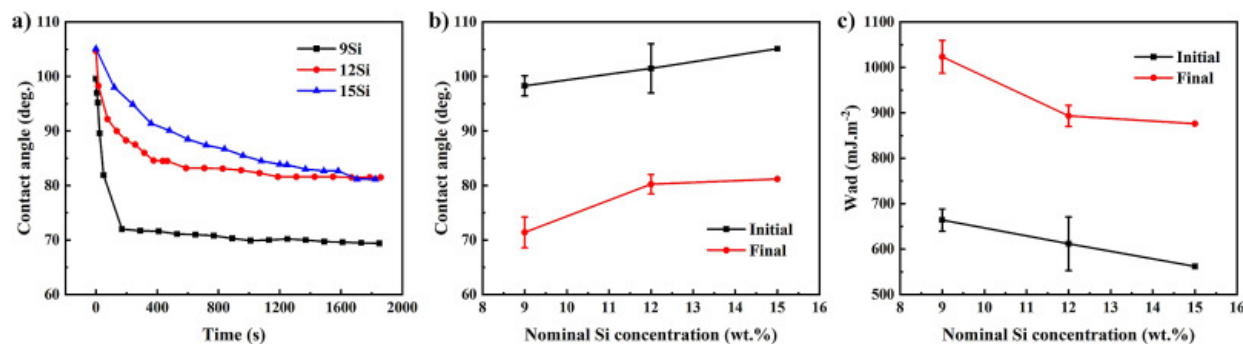
defined as the start of wetting. Following each experiment, the contact angle and droplet geometric parameters, such as contact diameter and height, were calculated using the Axisymmetric Drop Shape Analysis (ADSA) program.

After the wetting experiments, some samples were ground to remove the majority of the solidified droplet, and the residual alloy was etched away using a saturated NaOH solution. Selected samples were sectioned longitudinally and polished for microstructural characterization. Interfacial analyses were performed using optical microscopy (OM, Axio Imager A2m, Carl Zeiss, Germany) and scanning electron microscopy (SEM, VEGA 3 XMU, TESCAN, Czech) equipped with energy-dispersive X-ray spectroscopy (EDS). X-ray diffraction (XRD, Smartlab-9kW, Rigaku, Japan) was also employed to identify the interfacial phases.

3. Results

3.1. Al–Mg–Si/SiC system

[Fig. 2\(a\)](#) illustrates the evolution of the contact angle over time for Al–3Mg–xSi (x=9–15 wt%) alloys with varying Si content on SiC substrates at 1173 K. Increase Si content correlated with an increase in the initial contact angle, ranging from 98° for 9 wt% Si (Al–3Mg–9Si) to 105° for 15 wt% Si (Al–3Mg–15Si). A significant decrease in the spreading rate and an increase in the final contact angle (from 71° to 80°) were observed as the Si content increased from 9 wt% to 12 wt% ([Fig. 2\(b\)](#)). This coincided with a decrease in the work of adhesion from 1023 mJ/m² to 893 mJ/m² ([Fig. 2\(c\)](#)). Surface tension data used for these calculations were obtained by fitting the droplet images within the first 10 s of the wetting experiment to the Laplace equation using the ADSA software and are presented in [Table 1](#). These results indicate that even a small increase in Si content can negatively affect both wettability and adhesion.



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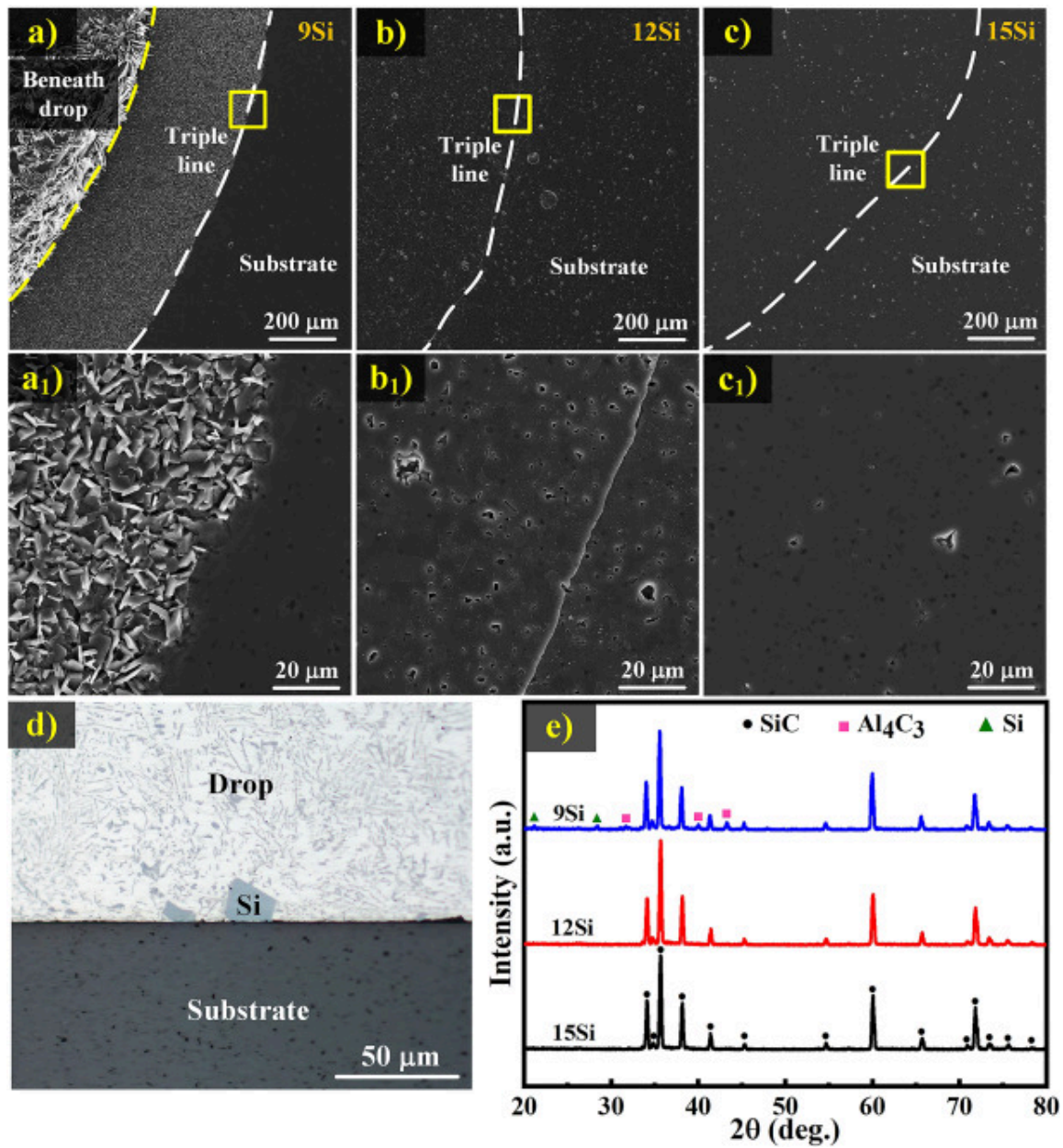
Fig. 2. (a) Contact angle variation with time for Al–3Mg–xSi alloys with varying Si contents on SiC substrates at 1173K; (b) initial and final contact angles, and (c) work of adhesion as a function of Si content.

Table 1. Measured surface tension values of Al–Mg–Si alloys in an Ar–10H₂ atmosphere.

Temperature (K)	Surface tension (mN·m ⁻¹)				
	Al–3Mg–9Si	Al–3Mg–12Si	Al–3Mg–15Si	Al–7Mg–12Si	Al–10Mg–12Si
1023	787	783	782	759	741
1073	783	778	778	757	732
1123	781	770	765	756	725
1173	776	764	760	752	709

Fig. 3 shows the formation of an Al₄C₃ phase at the interface for the Al–3Mg–9Si alloy, while this phase was absent at 12 wt%Si. This enhanced wettability observed with 9wt% Si is likely attributed to the formation of this Al₄C₃ interfacial layer. In the higher Si content systems, a small amount of blocky Si was observed at the solid–liquid interface

(Fig. 3(d)). This phase is believed to precipitate during solidification and was subsequently removed during etching with saturated NaOH solution, explaining its absence in the XRD analysis of the etched interface.

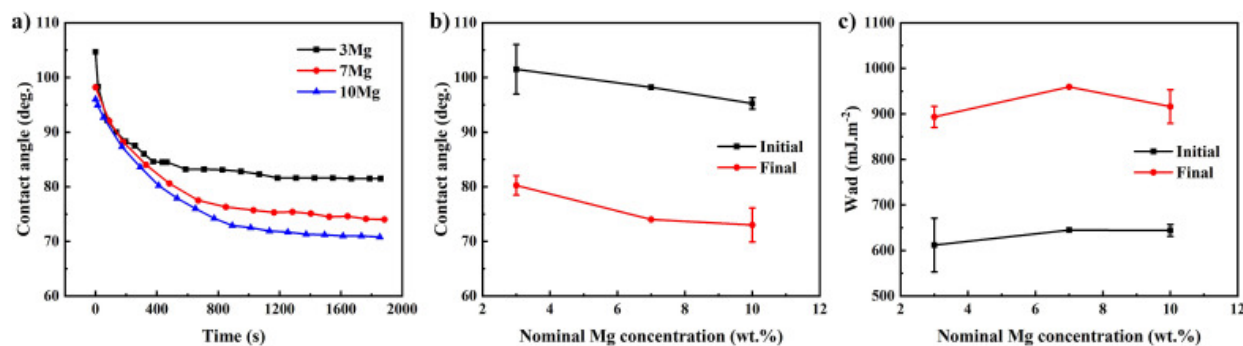


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Fig. 3. Microstructures at the triple line after 30min of isothermal wetting at 1173K, following removal of the solidified droplets, for Al–3Mg–xSi alloys with (a) 9, (b) 12, and (c) 15wt% Si on SiC substrates; (a₁–c₁) enlarged morphologies of the triple line regions; (d) cross-sectional morphology of the interfacial region after 30min of isothermal wetting of Al–3Mg–12Si on a SiC substrate; (e) XRD analysis of the corresponding regions in (a–c).

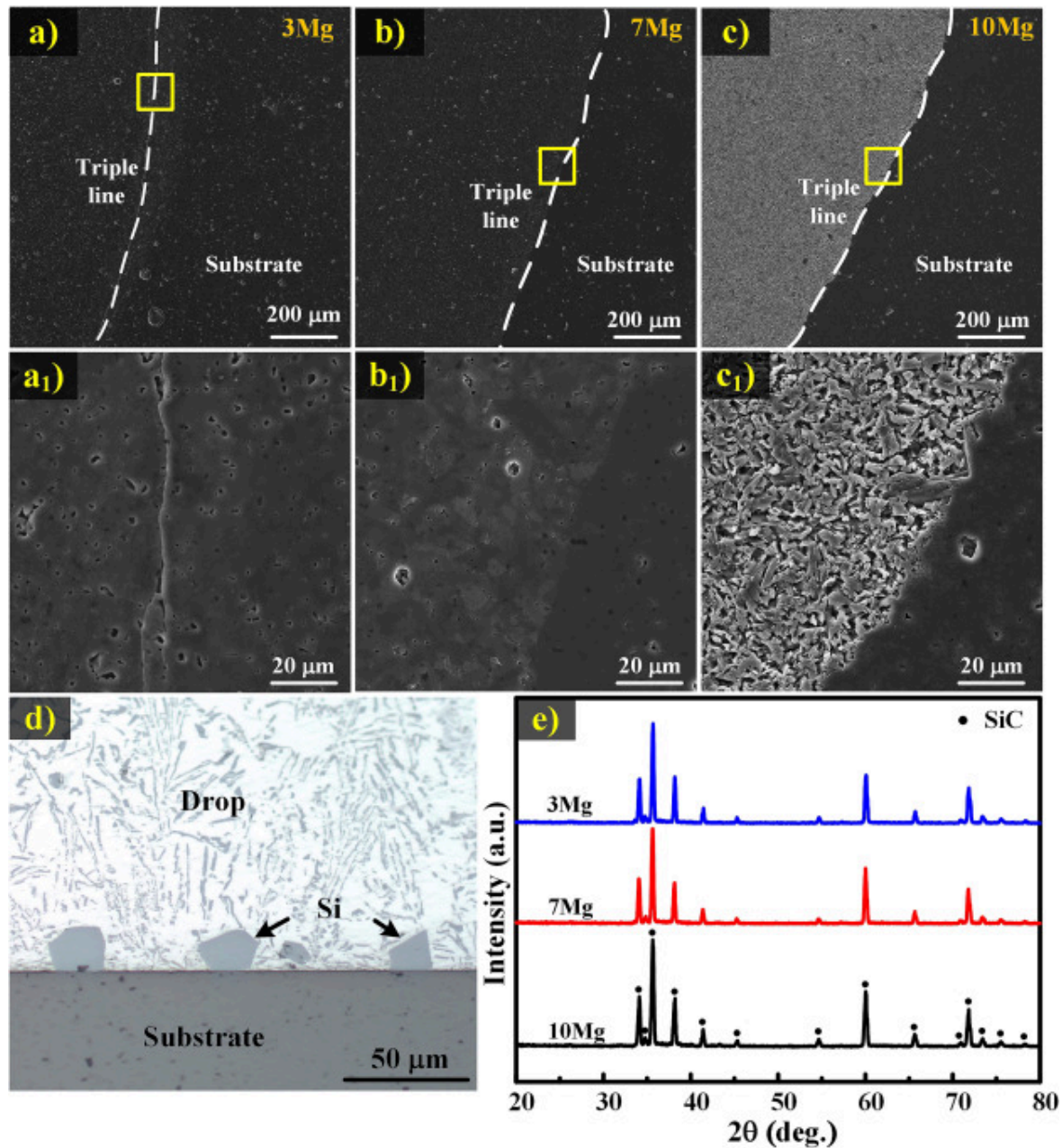
Fig. 4(a) presents the contact angle evolution for Al–xMg–12Si (x=3–10wt%) alloys with varying Mg content on SiC substrates at 1173K. Increasing Mg content led to a decrease in the final contact angle from 80° (3Mg) to 73° (10Mg), suggesting a slight improvement in wettability. Interestingly, despite Mg being a surface-active element known to reduce the surface tension of the alloy (Table 1), the final work of adhesion decreased from 959mJ·m⁻² (7Mg) to 916mJ·m⁻² (10Mg) at higher Mg contents. The interfacial microstructure of the triple-line region after 30min of wetting is shown in Fig. 5. Increasing the Mg content resulted in noticeable signs of high-temperature erosion in the interfacial region and an increase in the amount of blocky Si phase (Fig. 5(d)). However, EDS and XRD analyses did not reveal the formation of any new reaction products at the interface.



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Fig. 4. (a) Contact angle variation with time for Al–xMg–12Si alloys with varying Mg contents on SiC substrates at 1173K; (b) initial and final contact angles, and (c) work of adhesion as a function of Mg content.

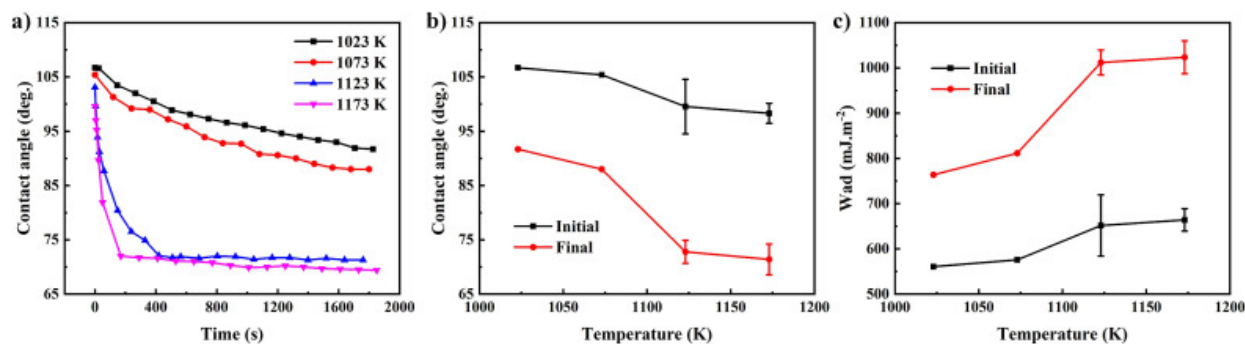


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Fig. 5. Microstructures at the triple line after 30min of isothermal wetting at 1173K, following removal of the solidified droplets, for Al–xMg–12Si alloys with (a) 3, (b) 7, and (c) 10wt%Mg on SiC substrates; (a₁–c₁) enlarged morphologies of the triple line regions; (d) cross-sectional morphology of the interfacial region after 30min of isothermal wetting of Al–7Mg–12Si on a SiC substrate; (e) XRD analysis of the corresponding regions in (a–c). In Fig. 5c₁, distinct corrosion marks rather than reaction products were observed on the substrate surface beneath the droplet. This unusual morphology is likely due to the high-temperature corrosion of the SiC substrate by the high-Mg-content Al–Mg–Si alloy.

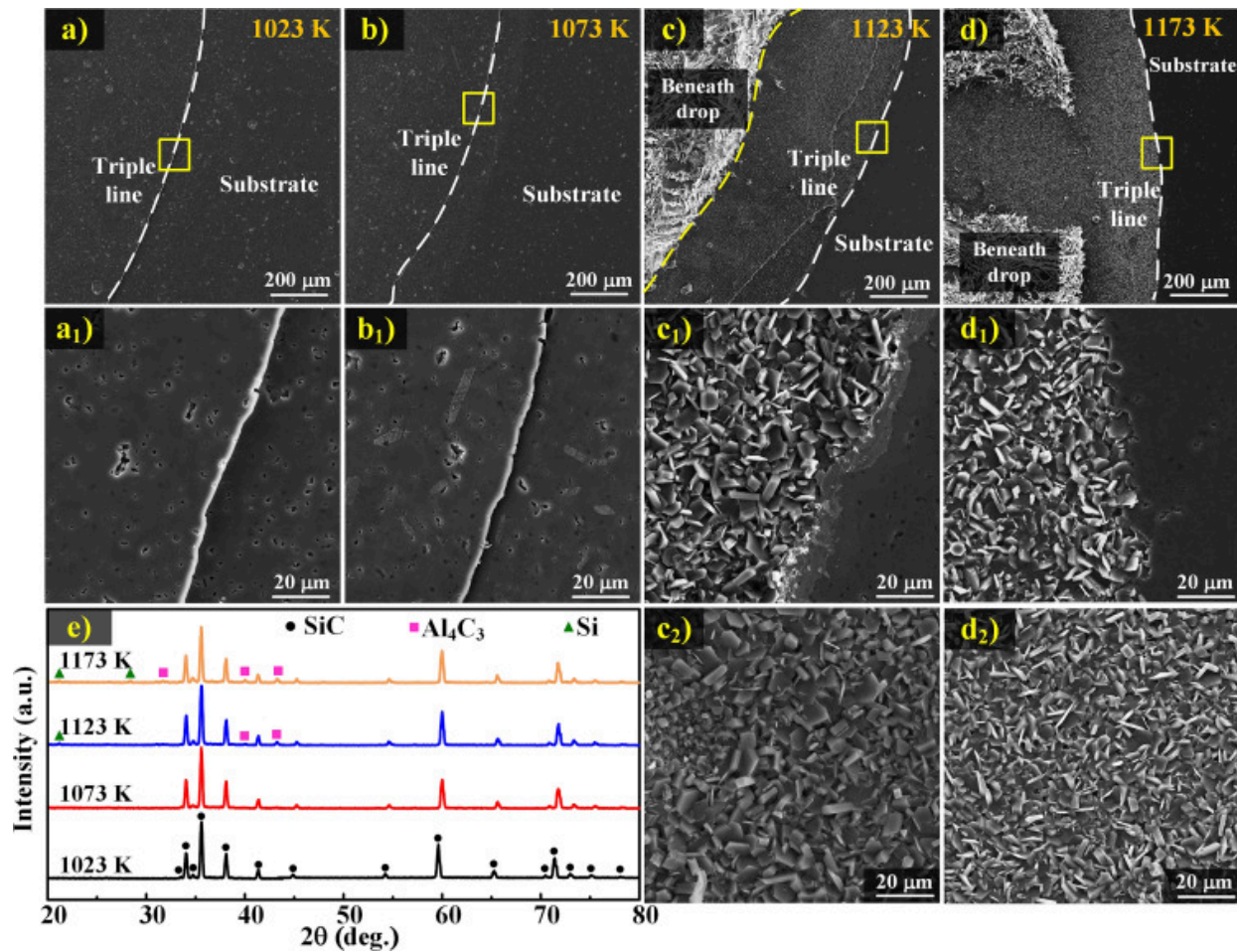
The effect of temperature on the wetting behavior of the Al–3Mg–9Si alloy on SiC substrates is presented in Fig. 6. Both wettability and spreading rate improved with increasing temperature, particularly between 1073K and 1123K. The final contact angle decreased from 88° (1073K) to 73° (1123 K), accompanied by an increase in the work of adhesion from 811 mJ·m^{–2} to 1012 mJ·m^{–2}. Fig. 7 shows the presence of abundant Al₄C₃ phase in the interfacial region at temperatures of 1223K and above. Similar to the influence of Si content, this improvement in wettability is primarily attributed to the formation of this reaction layer.



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Fig. 6. (a) Contact angle variation with time for Al–3Mg–9Si alloy on SiC substrates at 1023–1173K; (b) initial and final contact angles, and (c) work of adhesion as a function of temperature.



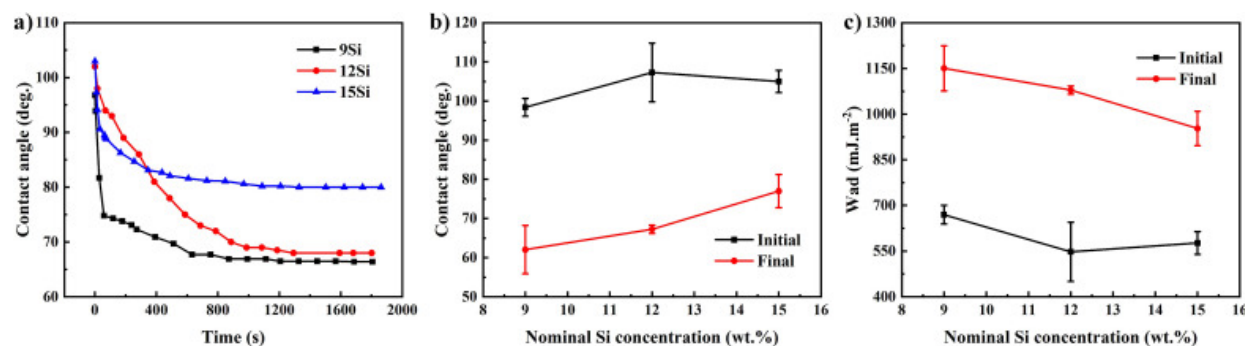
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Fig. 7. Microstructures at the triple line after 30min of isothermal wetting, following removal of the solidified droplets, for Al–3Mg–9Si alloy on SiC substrates at (a) 1023K, (b) 1073K, (c) 1123K, and (d) 1173K; (a₁–d₁) enlarged morphologies of the triple line regions; (c₂) and (d₂) morphologies of the central interfacial regions; (e) XRD analysis of the corresponding regions in (a–d).

3.2. Al–Mg–Si/SiO₂ system

The Al–Mg–Si/SiO₂ system exhibited a more pronounced reaction compared to the Al–Mg–Si/SiC system. Fig. 8(a) shows the contact angle evolution for Al–3Mg–xSi (x=9–15 wt%) alloys with varying Si content on SiO₂ substrates at 1073K. While the initial contact angle increased slightly from 98° (9Si) to 104° (15Si) with increasing Si content, the final contact angle showed a more substantial increase from 62° (9Si) to 77° (15Si) (Fig. 8(b)). This suggests that Si content has a limited effect on the initial wettability but significantly influences the final wettability in the Al–Mg–Si/SiO₂ system. The initial contact angles were comparable to those observed for the Al–Mg–Si/SiC system with the same alloy composition, indicating that SiO₂ does not inherently offer superior wettability compared to SiC. While the spreading rate and final contact angle appear improved on SiO₂ compared to SiC for the same alloy composition, this difference can be attributed to the varying reactivity and the nature of the reaction products formed in the two systems.



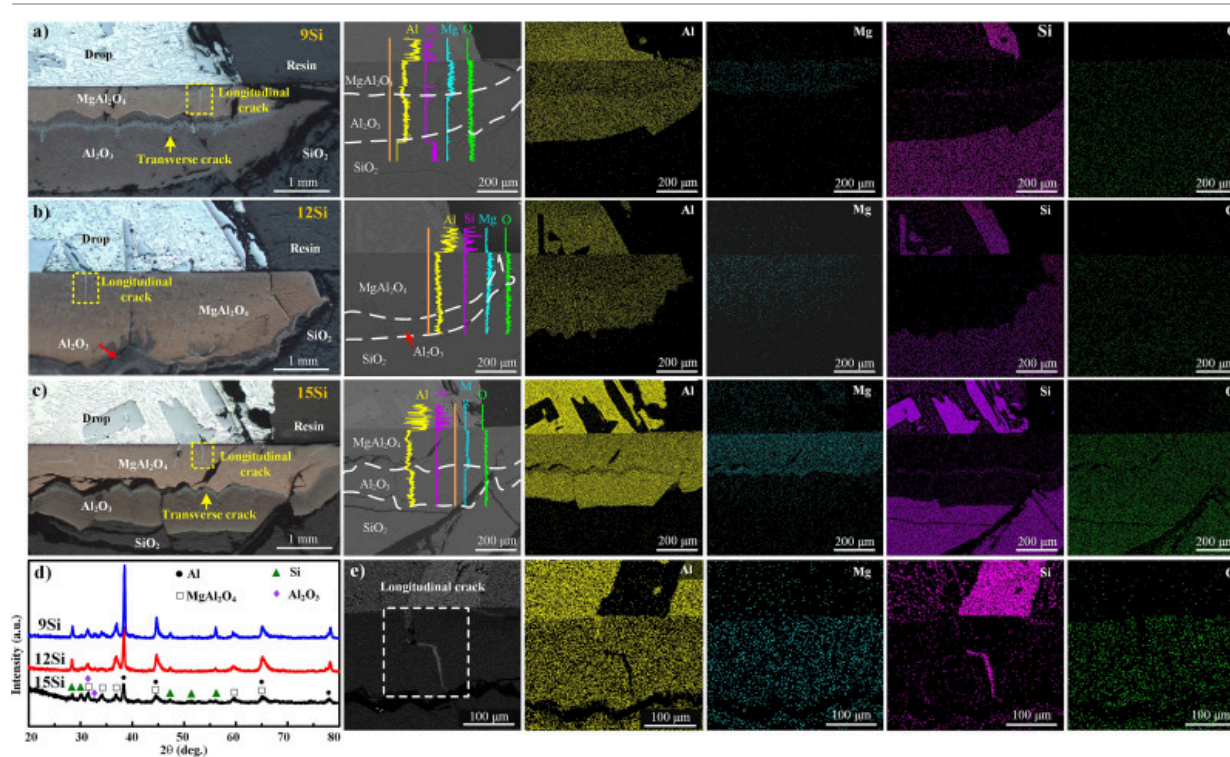
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Fig. 8. (a) Contact angle variation with time for Al–3Mg–xSi alloys with varying Si contents on SiO₂ substrates at 1073K; (b) initial and final contact angles, and (c) work of adhesion as a function of Si content.

The reaction layer in the Al–Mg–Si/SiO₂ system exhibited significant stratification. Combined EDS and XRD analyses (Fig. 9) identified MgAl₂O₄ adjacent to the droplet and Al₂O₃ closer to the unreacted substrate. The transformation of SiO₂ to either MgAl₂O₄ or

Al₂O₃ involves considerable volume shrinkage (theoretically, approximately 27% and 38%, respectively [26,27]). This shrinkage results in crack formation within the reaction layer. Longitudinal cracks were primarily filled with elemental Si, while transverse cracks contained both Si and Al₂O₃.



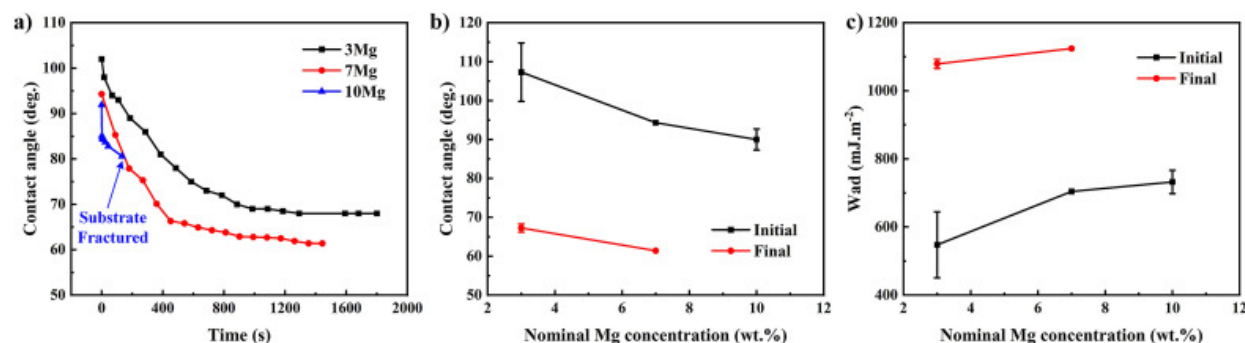
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Fig. 9. Cross-sectional morphologies and elemental distributions of the triple line regions after 30min of isothermal wetting at 1073K for Al–3Mg–xSi alloys with (a) 9, (b) 12, and (c) 15wt% Si on SiO₂ substrates; (d) XRD analysis of the reaction layer region; (e) enlarged image and elemental distribution of the longitudinal cracks in the reaction layer.

The influence of Mg on the wetting behavior is shown in Fig. 10(a), which depicts the contact angle evolution for Al–xMg–12Si (x=3–10wt%) alloys with varying Mg content on

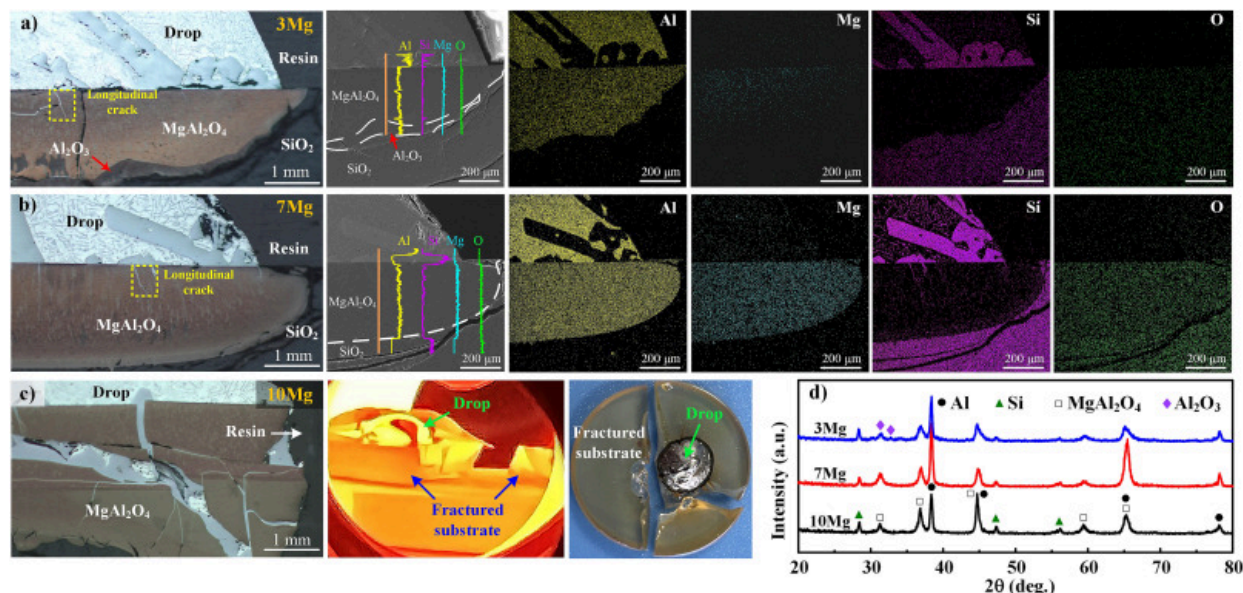
SiO₂ substrates at 1073K. Increasing Mg content decreased the initial contact angle from 107° (3Mg) to 90° (10Mg) and increased the initial work of adhesion from 548 mJ·m⁻² (3Mg) to 732 mJ·m⁻² (10Mg). This suggests that Mg enhances both initial wettability and adhesion, likely due to the strong affinity between Mg and SiO₂ [28]. The final contact angle also decreased from 67° (3Mg) to 61° (7Mg) with increasing Mg content, further indicating that Mg improves wettability after the reaction. However, at a nominal Mg content 10wt%, the substrate fractured readily after approximately 2 min of wetting due to the intense reaction between the droplet and the substrate (Fig. 11(c)), precluding the acquisition of complete contact angle data. Fig. 11 shows the cross-sectional morphology of the triple-line region after 30 min of wetting for Al–xMg–12Si alloys with different nominal Mg contents on SiO₂ substrates at 1073K. EDS and XRD analyses revealed a competitive formation between MgAl₂O₄ and Al₂O₃, with increasing Mg content favoring MgAl₂O₄ formation. At 7wt%Mg, Al₂O₃ was no longer detected.



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Fig. 10. (a) Contact angle variation with time for Al–xMg–12Si alloys with varying Mg content on SiO₂ substrates at 1073K; (b) initial and final contact angles, and (c) work of adhesion as a function of Mg content.



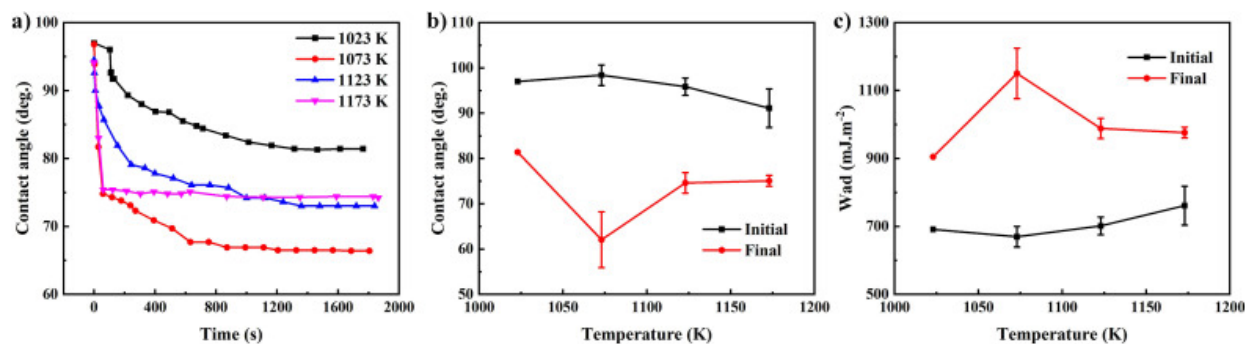
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Fig. 11. Cross-sectional morphologies and elemental distributions of the triple line regions after 30min of isothermal wetting at 1073K for Al–xMg–12Si alloys with (a) 3wt% and (b) 7wt%Mg on SiO₂ substrates; (c) fracture of the substrate after Al–10Mg–12Si droplet spreading on a SiO₂ substrate for 2 min; (d) XRD analysis of the reaction layer region.

Fig. 12 shows the effect of temperature on the wetting behavior of the Al–3Mg–9Si alloy on SiO₂ substrates. While the initial contact angle decreased only slightly from 97° (1023 K) to 91° (1173K) with increasing temperature, the spreading rate of the droplet increased significantly. Interestingly, the final contact angle exhibited a non-monotonic temperature dependence: it first decreased drastically from 81° (1023K) to 62° (1073K) and then gradually increased to 75° (1173K). Observation of the interfacial region (Fig. 13) revealed a similar trend in the thickness of the reaction layer, reaching a maximum at 1073K and decreasing thereafter. This behavior is possibly attributed to the accelerated interfacial reaction at higher temperatures, leading to rapid volume shrinkage of the

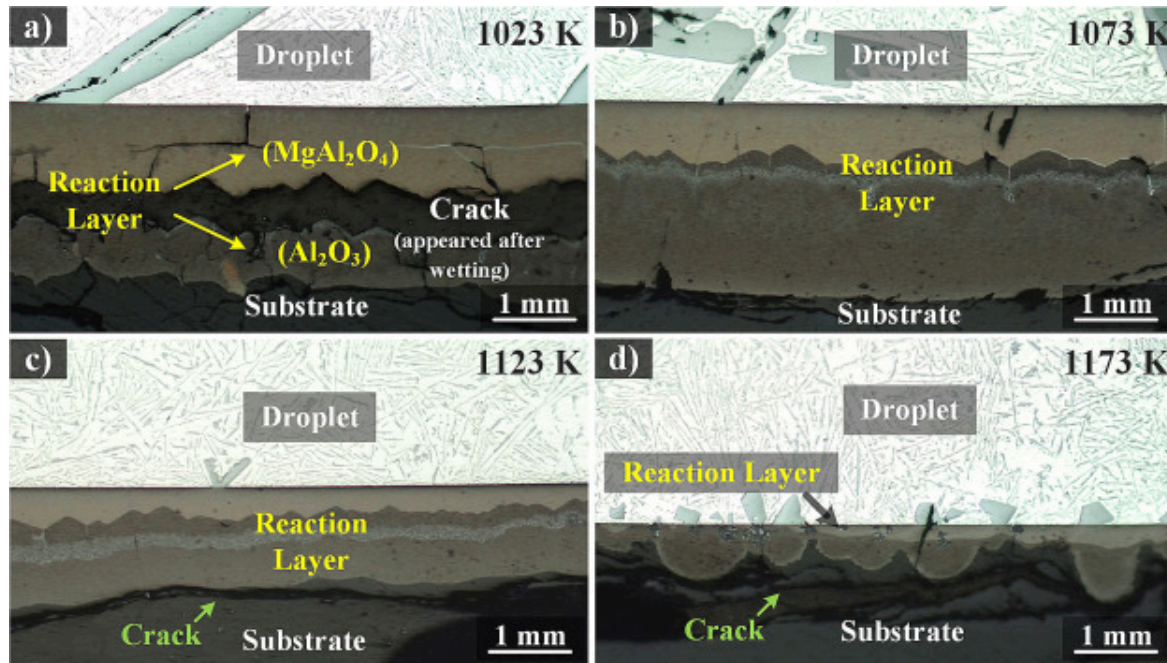
reaction layer due to phase transformation. This rapid shrinkage could generate excessive transient stress, causing fracture and hindering further reaction progress. This hypothesis is supported by the apparent fracture observed between the reaction layer and the substrate at temperatures of 1123 K and above (Fig. 13). However, unlike previous studies [23, 24], the formation of a mullite intermediate layer was not observed.



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Fig. 12. (a) Contact angles variation with time for Al–3Mg–9Si alloy on SiO₂ substrates at 1023–1173 K; (b) initial and final contact angles, and (c) work of adhesion as a function of temperature.



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Fig. 13. Cross-sectional morphologies of the interfacial regions after 30 min of isothermal wetting for Al–3Mg–9Si alloy on SiO₂ substrates at (a) 1023 K, (b) 1073 K, (c) 1123 K, and (d) 1173 K.

4. Discussion

4.1. Al–Mg–Si/SiC system

The dominant interfacial reaction in the Al–Mg–Si/SiC system is the formation of aluminum carbide (Al₄C₃), as shown in Eq. (3):

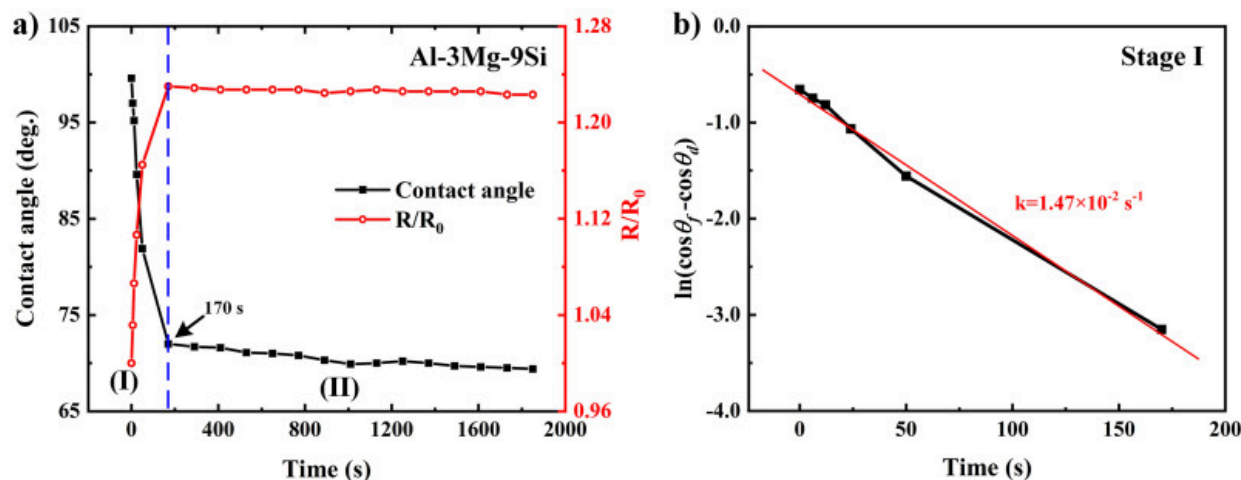


The silicon content in the alloy melt plays a critical role in regulating this reaction. Thermodynamic calculations by Lee et al. [29] suggest that a minimum Si concentration of approximately 10.6wt% is required to fully suppress Al₄C₃ formation at 1173K in the Al–Si/SiC system. This threshold demarcates two distinct wetting regimes. Above this critical Si concentration, the system is considered non-reactive, with spreading governed primarily by surface deoxidation processes. Conversely, below this threshold, the system becomes reactive, and spreading is primarily driven by Al₄C₃ formation at the interface. This observation is consistent with findings in other metal–ceramic systems, where reactive elements and the formation of wettable reaction products significantly influence the wetting behavior [4, 15, 29].

To analyze the spreading behavior of the Al–3Mg–9Si alloy on SiC at 1173K, we employed the reaction-controlled model proposed by Dezellus et al. [30],

$$\cos \theta_f - \cos \theta_d = \cos \theta_f - \cos \theta_0 \exp -kt, \quad (4)$$

where k is the spreading kinetic constant, t represents time, and θ_0 , θ_d , and θ_f are the initial, dynamic, and final contact angles, respectively. The wetting process can be divided into two stages (Fig. 14), with a clear transition at approximately 170s. The rapid spreading in the initial stage agrees well with the reaction-controlled model, suggesting that the formation of interfacial product is the dominant driving force.



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Fig. 14. (a) Variation of the contact angle and normalized droplet contact radius with time for the Al–3Mg–9Si alloy on a SiC substrate at 1173K; and (b) $\ln \cos \theta_f - \cos \theta_d$ vs. time during the first stage.

While magnesium does not directly participate in the interfacial reaction with SiC, its presence enhances wettability. Our previous study on the Al–Mg/SiC system [15] determined a critical Mg content of 19.4mol% for direct interfacial reaction at 1173K ($6\text{Mg} + 4\text{Al} + 3\text{SiC} = \text{Al}_4\text{C}_3 + 3\text{Mg}_2\text{Si}$). The addition of Si reduces the activity of Mg in Al–Mg–Si alloys [31], suggesting an even higher critical Mg concentration for direct reaction in the present system. Therefore, the observed improvement in wettability with increasing Mg content is likely attributed to three factors: (1) reduced droplet volume due to Mg evaporation, leading to a smaller apparent contact angle [32]; (2) decreased oxygen partial pressure around the droplet facilitated by Mg evaporation; and (3) the lower surface tension of Mg, contributing to a reduction in surface energy. However, given the intrinsic contact angle of pure Mg on SiC surface at 1173K is approximately 76° [15], further increases in Mg content would only marginally improve wettability.

4.2. Al–Mg–Si/SiO₂ system

The Al–Mg–Si/SiO₂ system exhibits a significantly more complex interfacial structure than the Al–Mg–Si/SiC system. The formation of various product phases is closely linked to the interfacial reactions, the penetration of Al and Mg into the substrate, and the diffusion of displaced Si atoms towards the droplet.

Upon contact, the SiO₂ dissolves, releasing [Si] and [O] into the melt (brackets denote dissolved species). The extremely low solubility of [O] in the alloy limits its diffusion, leading to the formation of an Al–Mg–Si–O liquid zone at the solid/liquid interface. Based on the Al–Mg–O phase diagram (Fig. 15(a)), the formation of MgAl₂O₄ and Al₂O₃ is primarily governed by the local Al and Mg concentrations. Furthermore, the lower surface tension of Mg compared to Al and Si in the Al–Mg–Si alloy ($T > 933\text{K}$), coupled with its strong affinity for SiO₂ [15], promotes preferential segregation of Mg at the alloy surface and interface. Given the higher reactivity of Mg compared to Al, we propose that [reaction \(5\)](#) occurs preferentially at the interface and triple-line region, resulting in volume shrinkage,

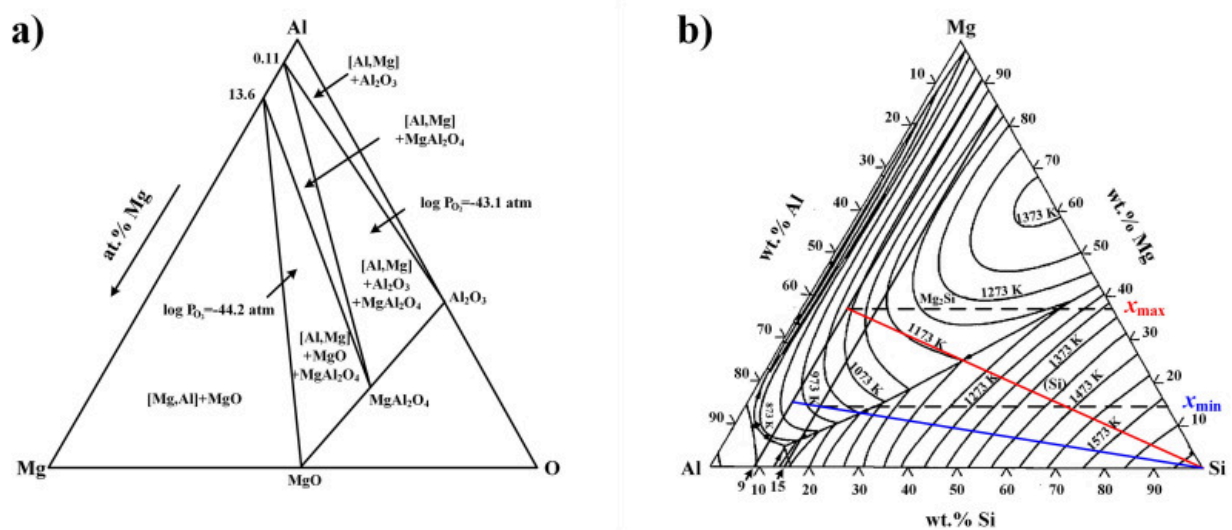


This is followed by the penetration or diffusion of the alloy droplet through the reaction layer, enabling further reaction with the substrate. This process continues until the Mg concentration falls below a critical threshold, after which [reaction \(6\)](#) becomes dominant, forming an Al₂O₃-rich reaction layer,



These interfacial reactions generate a substantial amount of [Si]. Assuming a constant [Al]/[Mg] ratio during [Si] accumulation, the Al–Mg–Si ternary phase diagram (Fig. 15(b)) suggests that approximately 14wt% (1023K, 15Si) to 37wt% (1173K, 9Si) of Mg is required for Mg₂Si formation under the experimental conditions. Therefore, we conclude that the supersaturated Si remains in its elemental form. Given its intermediate surface tension

between Mg and Al, and the rapid consumption of Mg in the initial reaction stage, we hypothesize that Si migrates towards the surface and interfacial regions.



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Fig. 15. (a) Al–Mg–O phase diagram [33] and (b) Al–Mg–Si phase diagram [34] at 1073 K. In (a), the point corresponding to an Mg content of 0.11 at.% is located very close to the Al apex. For clarity, this point has been slightly shifted along the Al–Mg binary line.

As noted, the interfacial reaction is accompanied by volume shrinkage and significant cracking, providing pathways for Si penetration and creating new solid/liquid interfaces for Si nucleation. In the deeper reaction layer regions, limited diffusion and the absence of longitudinal cracks (Fig. 9) hinder rapid Si transport to the droplet. Consequently, significant Si nucleation and growth occur on the surface of the solid product phases in these regions, explaining the substantial Si precipitation within the banded transverse cracks.

Si plays a multifaceted role in the alloy melt. As a product of (5), (6), its presence inherently limits their progression. Furthermore, Si addition reduces the activity of Mg in

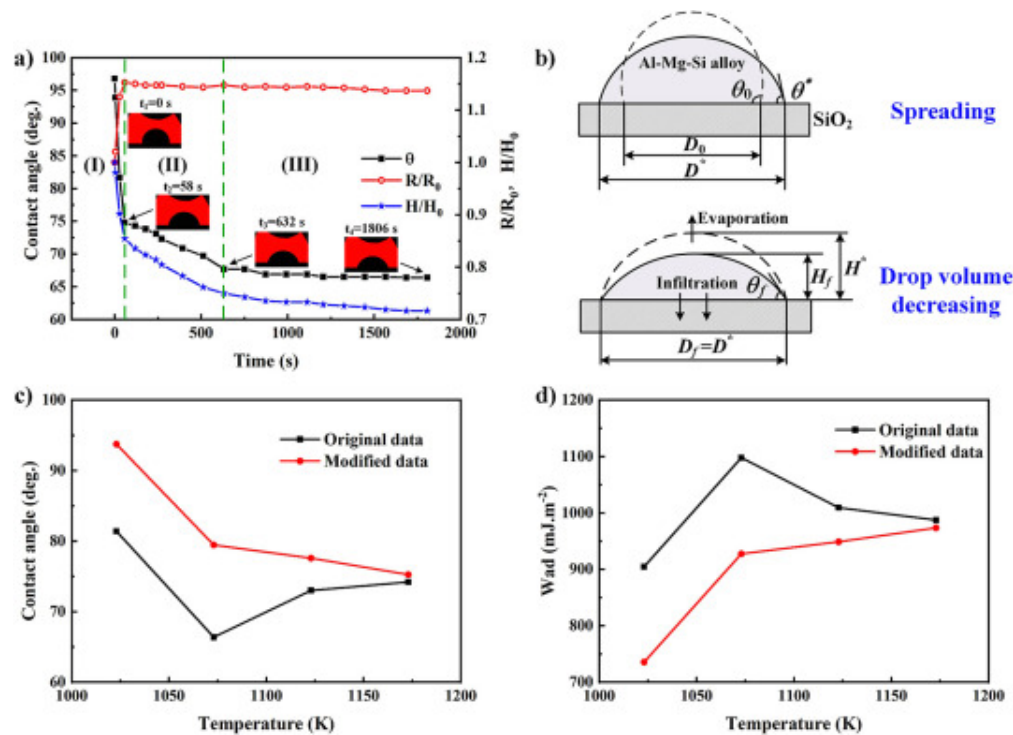
Al–Mg–Si alloys [31], potentially hindering reaction (5). However, our observations suggest that the overall influence of Si on reactivity is relatively minor.

Considering the Al–Mg–O phase diagram (Fig. 15(a)), MgO formation is thermodynamically possible with increasing Mg content. Shi et al. [35] investigated the phase transformations among MgO, MgAl₂O₄, and Al₂O₃ in the Al–Mg/SiO₂ system at 1073K, finding that reaction (7) proceeds spontaneously forward when the Mg content exceeds 9mol% (8.2wt%),



However, we did not detect MgO even at 10wt%Mg. This discrepancy likely arises from two factors: (1) the substantial Si content likely increases the Mg concentration required for MgO/MgAl₂O₄ equilibrium [31]; and (2) the preferential reaction of Mg with SiO₂ and its significant evaporation [32] drastically reduces the effective Mg concentration available for reaction (7).

Another crucial aspect is the significant alloy infiltration into the SiO₂ substrate. This infiltration, coupled with the reaction-induced volume reduction, affects the measured contact angle. Consequently, the observed contact angle should be considered an apparent contact angle. Relying solely on this apparent contact angle to evaluate wettability can be misleading. To illustrate, we examined the wetting behavior of Al–3Mg–9Si on SiO₂ at 1073K (Fig. 16(a)). The wetting process can be divided into three distinct stages: (I) rapid spreading, characterized by a rapid increase in contact diameter; (II) infiltration, where the droplet diameter remains constant while the height decreases; and (III) quasi-equilibrium, where both diameter and height remain relatively constant.



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Fig. 16. (a) Variation of the contact angle, normalized droplet contact radius, and normalized droplet height with time for the Al–3Mg–9Si alloy on a SiO₂ substrate at 1073K; (b) influence of volume changes, caused by droplet spreading and volume loss (due to infiltration and evaporation), on the apparent contact angle and droplet shape parameters; and (c–d) final contact angle and work of adhesion as a function of temperature for the Al–3Mg–9Si/SiO₂ system before and after correction.

In stage I, the reaction between the alloy droplet and SiO₂ drives rapid spreading and a significant decrease in the contact angle. Simultaneously, the alloy infiltrates the reaction layer, reducing droplet volume. Stage II, beginning at approximately 58s, is marked by arrested triple-line advancement, while the contact angle continues to decrease. This “pseudo-spreading” is attributed to ongoing infiltration and consequent droplet height

reduction. Finally, the system approaches wetting equilibrium in stage III, starting at approximately 632s. While Mg evaporation contributes to droplet volume reduction, its influence on the contact angle change is likely small due to the relatively low Mg content (3wt%) and its rapid consumption by reaction with SiO₂.

To account for volume reduction due to infiltration and evaporation, we employed the ideal spherical cap model proposed by Kondoh et al. [36], which compensates for the lost volume, enabling calculation of the true contact angle (θ^*) and a more accurate wettability assessment. Assuming a constant initial droplet volume (V_0) and a scenario where the droplet does not spread (Fig. 16(b)) while its volume decreases, maintaining a constant contact diameter ($D=D_0$), the droplet volume (V) during wetting can be expressed as:

$$V = \pi H^2 3D / \sin \theta - 2H / 6 \quad (8)$$

where H is the droplet height. The corrected droplet height (H^*) and radius (a^*) of the sphere containing the corrected spherical cap can be calculated from the following equations (For a detailed derivation, refer to [36]):

$$H^{*3} + 3D^2 H^* / 4 - 6V_0 / \pi = 0 \quad (9)$$

$$a^* = V_0 / \pi H^{*2} + H^* / 3 \quad (10)$$

Finally, the corrected contact angle (θ^*) is determined using:

$$\theta^* = \sin^{-1} D / 2a^* \quad (11)$$

Fig. 16(c) presents the corrected contact angle and work of adhesion as a function of temperature for the Al–3Mg–9Si/SiO₂ system. Correcting for droplet volume change eliminates the non-monotonic temperature dependence observed in the apparent contact angle (Fig. 12). The corrected data reveal a clear trend: both wettability and work of adhesion improve with increasing temperature. This highlights the importance of

accounting for droplet volume variations when analyzing wetting behavior, particularly in reactive infiltration systems.

5. Conclusions

This study investigated the wetting behavior of Al–Mg–Si alloys on SiC and SiO₂ substrates in an Ar–10%H₂ atmosphere using a dispensed sessile drop method, focusing on the effects of alloying element concentrations and temperature on wettability and interfacial microstructure. The key findings are:

- (1) On SiC substrates, increasing temperature and decreasing Si content enhance Al₄C₃ formation and wettability. Mg indirectly improves wettability by reducing the oxygen partial pressure and surface tension. On SiO₂ substrates, increased Mg content improves wettability, while Si content has a less pronounced effect.
- (2) Interfacial reactions on SiO₂ produce a layered structure characterized by MgAl₂O₄ and Al₂O₃ formation, driven by melt infiltration and Si diffusion. Volume shrinkage within this interfacial layer leads to crack formation.
- (3) The apparent non-monotonic temperature dependence of the final contact angle of Al–Mg–Si alloys on SiO₂, an artifact rising from volume reduction due to droplet infiltration and evaporation, was corrected using an ideal spherical cap model.
- (4) Pre-oxidation of SiC primarily controls Al₄C₃ formation rather than directly enhancing wettability. This finding has significant implications for the fabrication of SiC-reinforced AMC.

CRedit authorship contribution statement

Yu-Jie Ge: Writing – original draft, Investigation, Formal analysis, Data curation. **Rui-Fen Guo:** Formal analysis. **Ping Shen:** Writing – review & editing, Supervision, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work is supported by National Key Research and Development Program of China (No. [2022YFB3705701](#)) and the National Natural Science Foundation of China (No. [52071155](#)).

Appendix A. Supplementary data

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Supplementary material: Wetting of SiC and SiCox substrates by molten Al and its alloys.

Data availability

The primary raw/processed data required to reproduce these findings have been indicated in this article. The others cannot be shared at this time as they also form part of an ongoing study.

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